Falcon Fixed Target

An explanatory report

The question. A lattice signer hashes a message with a fresh random salt, samples a candidate signature, and rejects candidates whose norm is too large. Can it retain that same salt and hash target while retrying, and still admit a concrete security proof?

The answer in the declared model. A discrete Gaussian with Falcon's original width remains sufficiently uniform across target cosets after the exact integer norm cutoff. At the two parameter sets, the acceptance-probability variance is below 2^{-90} and 2^{-120}, and the accepted public image has chi-square distance below 2^{-81} and 2^{-83} from uniform. These analytical inequalities, evaluated with rigorous numerical enclosures, provide the missing distributional input to a fixed-target CoreFalcon security reduction.

The reduction retains explicit assumptions about the key distribution and exact-arithmetic sampler, as well as the costs of random-oracle programming, forgery extraction, and simulation. The primary concrete application uses a classical random oracle. The quantum-oracle corollary carries additional quantitative costs. The report develops the mathematical objects before using them, follows the central estimates, and then assembles the security statement and its scope.

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1 Why a retry policy becomes a security question

A digital signature lets a recipient check that a message was approved by someone holding a secret key. Producing the signature should require that secret, while checking it should require only the corresponding public key. In the lattice signature studied here, the signer must find a short integer vector satisfying a public equation. Its secret key makes a suitable sampling procedure efficient. A verifier checks the equation and the vector's length.

Sampling does not always produce a short enough vector. The signer therefore rejects an unsuccessful sample and tries again. The question is what remains fixed between attempts. It can keep the original equation and sample another solution, or change the equation's target before resampling. Both procedures can return perfectly valid signatures. Nevertheless, their output distributions need not agree, and a security argument for one procedure need not prove security of the other.

Suppose two targets are initially equally likely. At target 0 a proposed solution passes with probability $1/2$; at target 1 it always passes. If the target is chosen once and retained, target 0 merely takes longer to finish. The final target probabilities remain $(1/2, 1/2)$. If every failed attempt restarts with a new uniform target, the successful masses in one attempt are $(1/4, 1/2)$. Normalizing gives final probabilities $(1/3, 2/3)$. The second procedure favors the easier target.

This distinction matters because a security proof often answers signing requests through a simulator that has no secret key. Rather than solve a chosen equation, the simulator can sample a short vector first and calculate which target it solves. It then arranges the modeled hash value to equal that target. For this strategy to be useful, its joint distribution of targets and vectors must be sufficiently close to the real signer's distribution. Returning valid vectors alone is insufficient.

The central result controls this distributional comparison after Falcon's norm test. It applies at the original parameters, uniformly over keys satisfying the stated geometric promises. The resulting security statement concerns the mathematical CoreFalcon model: a fresh initial salt, one target per signing request, and private retries at that target. The proof has an analytic part, which studies a norm-conditioned integer Gaussian, and a cryptographic part, which converts that estimate into a simulation and forgery reduction. We first identify exactly which probability laws these two parts must connect.

2 The public equation and its Gaussian solutions

2.1 A target selects a set of integer vectors

Fix $q = 12289$ and $n \in \{512, 1024\}$. A ring element is represented by a polynomial of degree less than n. Multiplication is performed with the rule $X^n = -1$. The integer-coefficient ring and its finite reduction modulo q are

$$\mathcal{R} = \mathbb{Z}[X]/(X^n + 1), \qquad \mathcal{R}_q = \mathcal{R}/q\mathcal{R}.$$

The public key is $h \in \mathcal{R}_q$. A proposed solution is a pair $z = (u, v) \in \mathcal{R}^2$, with public target, or *syndrome*,

$$F_h(u, v) = u + hv \pmod{q}.$$

Writing the coefficients of u and v consecutively identifies z with a vector in $\mathbb{Z}^d$, where $d = 2n$. Its squared norm is the sum of the squares of all $2n$ coefficients. Every norm of a signature vector uses this coefficient representation.

The solutions of $F_h(z) = 0$ form a lattice $\Lambda_h \subset \mathbb{Z}^d$: a discrete additive set closed under integer linear combinations. If z_y is one solution of $F_h(z) = y$, all solutions at that target are precisely

$$F_h^{-1}(y) = z_y + \Lambda_h.$$

Such a translated lattice is called a *coset*. Subtracting two solutions at the same target gives a lattice vector; adding a lattice vector to one solution gives another. A target therefore specifies a whole collection of possible vectors, not a unique answer.

The map is onto: for any y, the pair $(u, v) = (y, 0)$ is a solution modulo q. This does not usually give a short enough solution to sign. There are q^n different targets, so the quotient

$$Y = \mathbb{Z}^d/\Lambda_h$$

has cardinality $|Y| = q^n$ and can be identified with $\mathcal{R}_q$. This algebraic fact holds for every public h; it does not require h to be invertible. Let $U(y) = 1/|Y|$ denote the uniform target law.

2.2 A discrete Gaussian and its conditional law

A Gaussian on integer vectors assigns greater probability to shorter vectors. For width $\sigma > 0$, define

$$G(z) = \frac{\exp(-\|z\|^2/(2\sigma^2))}{\sum_{x\in\mathbb{Z}^d} \exp(-\|x\|^2/(2\sigma^2))}.$$

The denominator normalizes the positive masses to sum to one. This is an integer distribution, even though its weights use the familiar Gaussian exponential. The width specifies the exponent; the variance of a discrete coordinate is not defined to equal σ^2.

Let

$$\mu(y) = \Pr_{Z\leftarrow G}[F_h(Z) = y], \qquad G_y(z) = \frac{G(z)}{\mu(y)}\,\mathbf{1}[F_h(z) = y].$$

Here G_y is the Gaussian restricted to the solutions at target y and renormalized. The public experiment draws Z from G and then computes $F_h(Z)$. A reverse experiment first chooses y and then draws from G_y. If the reverse experiment chooses y according to μ, their joint laws agree exactly. If it chooses a uniform y, the agreement depends on how close μ is to uniform.

This is the role of Gaussian *smoothing*: at a sufficiently large width, the total Gaussian masses of different lattice cosets become nearly equal. The secret basis controls a geometric quantity that permits a uniform smoothing estimate, rather than an estimate based on a typical sampled key. The next part of the report establishes that geometric input. Smoothing before any rejection, however, is only a starting point. Restricting to short vectors can favor some targets over others.

3 What the signer does and what the verifier sees

The accepted event is

$$A = \{z : \|z\|^2 \le T\}.$$

Because squared norms of integer vectors are integers, rejection starts at $K = T + 1$, not at an unspecified real boundary.

The relevant parameters are

	Degree 512	Degree 1024
Coefficient dimension d	1024	2048
Gaussian width σ (approximately)	165.736617	168.388571
Largest accepted squared norm T	34,034,726	70,265,242
First rejected squared norm K	34,034,727	70,265,243
Salt length κ	320 bits	320 bits

The exact width formula will be given with the geometric and numerical estimates. Rounded decimals are descriptive rather than definitions.

The hash is modeled as a random oracle

$$H : \mathsf{PK} \times \{0,1\}^{\kappa} \times \mathsf{M} \longrightarrow \mathcal{R}_q.$$

This means that distinct inputs have independent uniform outputs, while repeating an input returns its previously assigned value. The tuple contains the public key, salt and message with an unambiguous encoding. The model concerns one honestly generated key. It does not assert that an ordinary implemented hash can literally be programmed.

For a message m, the signer draws an independent uniform salt r after receiving the message and sets $y = H(h, r, m)$. It then repeatedly draws (u, v) from its secret-key sampling distribution P_y, using fresh private coins, until A holds. Throughout this loop it retains both r and y. It returns the mathematical signature (r, v).

The declared exact-arithmetic fast-Fourier sampler has law P_y, which is not assumed identical to G_y. Its proved probability-ratio profile enters the security comparison in Section 7.2; Section 5.2 checks the geometric hypotheses needed there.

Verification recomputes y and reconstructs

$$\widehat{u} = \mathrm{ctr}_q(y - hv),$$

where ctr_q reduces every coefficient to an integer in $[-6144, 6144]$. It accepts the mathematical signature when $\|(\widehat{u}, v)\|^2 \le T$. Centering cannot increase any coefficient's absolute value, so an accepted sampled pair always yields a valid signature. The sampled u can differ from $\widehat{u}$, especially at degree 1024. The probability comparison therefore studies the sampled pair before this public reconstruction.

Only the final signature, or a specified public error, is revealed. Retry counts, intermediate vectors and running times are absent from the interface. Messages may be chosen adaptively using previous answers. Signing requests remain classical even in the quantum-oracle variant; quantum access concerns the hash queries. A fresh random salt makes a previously queried signing input unlikely, but does not make collisions impossible. The security proof pays for this issue explicitly before replacing each signing target by a fresh uniform one.

These conditions define the signing interface used below. Section 6 treats the canonical compressed wrapper, and Section 8.2 collects changes requiring further analysis.

4 Three distributions of accepted targets

Write

$$a_y = \Pr_{Z \leftarrow G_y}[A], \qquad \delta = \Pr_{Z \leftarrow G}[A^c].$$

The first number depends on the target; the second averages rejection under the ambient Gaussian. Assume for now that every a_y is positive, as the later uniform rejection bound will establish.

4.1 Choose a uniform target and keep it

In the ideal uniform-target experiment, choose $y \leftarrow U$ once and repeat independent G_y draws until acceptance. The accepted vector law at that target is $G_y \mid A$. Indeed, the probability of eventually returning a particular accepted vector z is

$$\sum_{j=0}^{\infty}(1-a_y)^j G_y(z) = \frac{G_y(z)}{a_y}.$$

Every chosen target eventually succeeds, so its marginal law remains U. In the proof experiment where each signing request receives a fresh uniform target, the real sampler analog is the joint law

$$L(y,z) = U(y)P_y(z \mid A).$$

The norm theorem first addresses the ideal Gaussian; the sampler comparison then returns to this real accepted law.

4.2 Choose a new uniform target after each failure

Suppose instead that each attempt chooses a new uniform target and one G_y sample. The successful target masses per attempt are $U(y)a_y$. The final target law is therefore

$$\widetilde{\nu}(y) = \frac{U(y)a_y}{\mathbb{E}_{y\leftarrow U}a_y}.$$

Targets with higher acceptance probability appear more often. This is a useful comparison experiment, but it is not yet the public simulator's experiment.

4.3 Accept an ambient Gaussian and compute its target

The public simulator draws from G until the norm test succeeds and then computes the syndrome. It knows no trapdoor. The probability that one ambient draw both has target y and passes is $\mu(y)a_y$. Hence its accepted target law and joint law are

$$\nu(y) = \frac{\mu(y)a_y}{1-\delta}, \qquad J(y,z) = \nu(y)G_y(z \mid A).$$

Equivalently, J samples $z \leftarrow G \mid A$ and sets $y = F_h(z)$. The conditional ideal vector law agrees with fixed-target signing, but the target weights differ. Both the unconditioned image $\mu(y)$ and the acceptance factor a_y must be controlled.

4.4 A finite example showing all three laws

Consider a public proposal with three possible outcomes:

Outcome	Target	Passes?	Probability
z_0^+	0	Yes	3/8
z_0^-	0	No	3/8
z_1^+	1	Yes	1/4

Its target distribution is $\mu = (3/4, 1/4)$, while the conditional acceptance probabilities are $(a_0, a_1) = (1/2, 1)$. Keeping an initially uniform target produces $(1/2, 1/2)$. Redrawing a uniform target produces

$$\widetilde{\nu} = \frac{(1/4, 1/2)}{3/4} = (1/3, 2/3).$$

Accepting the public proposal produces

$$\nu = \frac{(3/8, 1/4)}{5/8} = (3/5, 2/5).$$

All procedures return a passing outcome, yet all three target laws differ. The distinction also changes expected attempt counts: respectively $3/2$, $4/3$ and $8/5$. These are illustrative probability calculations, not small Falcon instances or speed estimates.

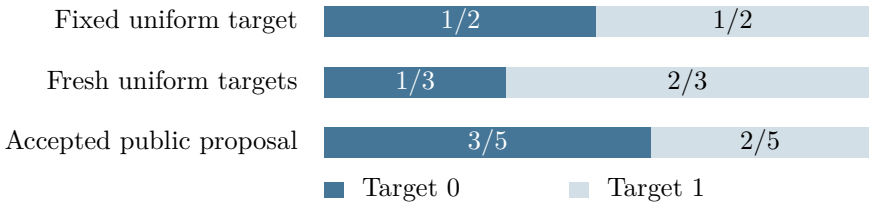


Figure 1: The three accepted target laws in the finite example. Every row is a probability distribution summing to one. The public proposal has target weights $(3/4, 1/4)$ before acceptance; the two target acceptance probabilities are $(1/2, 1)$.

The mathematical task can now be stated precisely. Ordinary smoothing controls the ambient target weights. The new norm-tail analysis must control the effect of keeping only short vectors. Together, these estimates must make the accepted public law sufficiently close to uniform-target fixed retries, in the direction needed by the security reduction. A small average bias alone will not be the entire argument: the proof must also rule out accepted targets with dangerously tiny probability and include the actual sampler error.

5 Controlling a sharp integer norm cutoff

We now bound the accepted ambient image just identified, uniformly over the permitted keys. The proof separates ordinary Gaussian smoothing from the additional bias introduced by the integer norm cutoff.

5.1 The precise geometric statement

Let n be either 512 or 1024, let $d = 2n$, and set $q = 12289$ and $B = 1.17\sqrt{q}$. Consider any full-rank sublattice $\Lambda \subset \mathbb{Z}^d$ of index $N = q^n$ that has a basis whose Gram–Schmidt lengths are all at most B. The index is the number of cosets: the quotient group $Y = \mathbb{Z}^d/\Lambda$ has N elements. The theorem does not require a typical key or an average over key generation.

An ordered basis $b_1, \ldots, b_d$ generates the lattice by integer linear combinations. Its Gram–Schmidt vector $\widetilde{b}_i$ is the part of b_i orthogonal to the real span of the preceding basis vectors. Bounding these orthogonal lengths controls Gaussian sums even if the original basis vectors themselves are longer.

Write $X \sim G$ for an ambient Gaussian vector, in product form:

$$\Pr[X = x] = \frac{e^{-\alpha\|x\|^2}}{Z_\alpha^d}, \qquad \alpha = \frac{1}{2\sigma^2}, \qquad Z_\alpha = \sum_{j\in\mathbb{Z}} e^{-\alpha j^2}. \tag{5.1}$$

The coordinates are independent before conditioning and are dimensionless integers. The admissible parameters are the following exact decimal intervals and integer cutoffs.

n	d	Width interval for σ	T
512	1024	[165.736617182977, 165.736617182979]	34034726
1024	2048	[168.388571446543, 168.388571446545]	70265242

For a coset $y \in Y$, define its acceptance probability by

$$a_y = \Pr[\|X\|^2 \le T \mid X \bmod \Lambda = y].$$

Let U be uniform on Y, and let ν be the law of $X \bmod \Lambda$ conditional on $\|X\|^2 \le T$. For every lattice and every width in the corresponding interval, the result is

	$n = 512$	$n = 1024$
$\mathrm{Var}_{y\sim U}(a_y)$	$< 2^{-90}$	$< 2^{-120}$
$\chi^2(\nu\|U)$	$< 2^{-81}$	$< 2^{-83}$.

(5.2)

Here variance measures variation over uniformly chosen targets, and $\chi^2(\nu\|U) = \mathbb{E}_U[(d\nu/dU - 1)^2]$ measures the mean squared departure of the accepted syndrome density from one. Neither statement identifies the overall rejection probability with these tiny numbers. The same bounds remain valid if acceptance also requires $|X_j| \le 2047$ on any fixed set of n coordinates.

5.2 Why the key-quality check supplies the basis hypothesis

The public map is $F_h(s_1, s_2) = s_1 + hs_2 \pmod q$, with $h = g/f \pmod q$ in $\mathbb{Z}_q[X]/(X^n + 1)$ and f invertible modulo q. The secret polynomials satisfy $fG - gF = q$. Since the first component of F_h has coefficient one, the map is onto and $\Lambda_h = \ker F_h$ has index q^n.

Expand the two polynomial columns $u_1 = (g, -f)$ and $u_2 = (G, -F)$ into all their negacyclic coefficient shifts, placing the first block before the second. Their coefficient determinant has absolute value q^n, so they generate exactly Λ_h. A negacyclic shift rotates coefficients with a possible sign change and preserves Euclidean length.

Write $a^* = a(X^{-1})$ in the quotient ring and $D = ff^* + gg^*$. The fractions below are computed in the real coefficient algebra $\mathbb{R}[X]/(X^n+1)$. At a complex root ξ of X^n+1, $D(\xi) = |f(\xi)|^2+|g(\xi)|^2 > 0$: the equation $fG - gF = q$ prevents f and g from vanishing together. Thus D is invertible. The mathematical key-generation check is

$$\max\left\{\|(g,-f)\|, \left\|\left(\frac{qf^*}{D}, \frac{qg^*}{D}\right)\right\|\right\} \le B.$$

This initially bounds two polynomial vectors. To see why it bounds every ordinary Gram–Schmidt length, let W be the real span of the shifts of u_1. Orthogonal projection cannot increase a length, so all first-block Gram–Schmidt vectors have length at most $\|u_1\|$. For the second block, the identity

$$u_2 - \frac{Gg^* + Ff^*}{D}u_1 = v := \left(\frac{qf^*}{D}, \frac{qg^*}{D}\right)$$

exhibits the projection onto $W^{\perp}$: the polynomial inner product of u_1 and v vanishes, and therefore v is orthogonal to every shift of u_1. Shifts preserve W and its orthogonal complement. Each second-block vector first projects to a shift of v; further Gram–Schmidt projections can only shorten it. Hence every resulting length is at most B. The argument permits any ordering within a block.

Splitting a polynomial as $a(X) = a_{\text{even}}(X^2) + X a_{\text{odd}}(X^2)$ merely rearranges its coefficients into the even and odd positions, preserving their total squared norm. Repeating this split expands the sampling tree into ordinary coefficient coordinates. The exact tree factorization identifies the sampler's unnormalized leaf values D_i with these squared lengths. Thus $\sqrt{D_i} \le B$ and its one-dimensional widths satisfy $\sigma_i = \sigma/\sqrt{D_i} \ge \sigma/B$. This establishes the geometric input to the exact-arithmetic sampler profile. It is separate from proving that profile, or validating finite-precision arithmetic.

The exact width is determined by

$$\sigma = \frac{B}{\pi}\sqrt{\frac{\log(4n(1+1/\varepsilon_{\text{des}}))}{2}}, \qquad \varepsilon_{\text{des}} = \frac{1}{\sqrt{2^{64}\lambda}}, \qquad T = \left\lfloor 1.21 d\sigma^2 \right\rfloor,$$

where log is natural and $\lambda = 128$ or 256, respectively. The parameter λ specifies the width design; it is not a proved bit security level. These exact widths lie in the displayed intervals.

5.3 Separate the original image from the rejected image

Since squared norms are integers, rejection starts at $K = T + 1$. Introduce two densities relative to U:

$$p(y) = N \Pr[X \bmod \Lambda = y], \qquad r(y) = N \Pr[X \bmod \Lambda = y,\ \|X\|^2 \ge K].$$

Their uniform means are $\mathbb{E}_U p = 1$ and $\mathbb{E}_U r = \delta = \Pr[\|X\|^2 \ge K]$. The conditional rejection probability at a particular target is $\delta_y = r(y)/p(y)$, so $a_y = 1 - \delta_y$. Positivity of every coset Gaussian mass makes this division well-defined. The accepted image has the exact density

$$\frac{d\nu}{dU}(y) = \frac{p(y) - r(y)}{1 - \delta}. \tag{5.3}$$

The normalization in this formula is ambient acceptance, not the uniform-target average of a_y.

The two centered norms to control are

$$Q_0 = \left(\mathbb{E}_U |p - 1|^2\right)^{1/2}, \qquad R = \left(\mathbb{E}_U |r - \delta|^2\right)^{1/2}.$$

Ordinary smoothing controls Q_0. The new difficulty is R, because the indicator of a sharp integer norm cutoff does not factor into independent coordinate weights.

For $c > 0$, write $\rho_c(S) = \sum_{x \in S} e^{-c\|x\|^2}$ and $\vartheta(c) = \sum_{j \in \mathbb{Z}} e^{-cj^2}$. The dual lattice $\Lambda^* = \{w : \langle w, \Lambda \rangle \subset \mathbb{Z}\}$ represents the frequencies of the quotient: its characters are $\widehat{Y} = \Lambda^*/\mathbb{Z}^d$. A character evaluates the phase $e^{2\pi i \langle w, x \rangle}$ and is constant on each coset. Finite Fourier orthogonality gives Parseval's identity

$$R^2 = \sum_{w \in \widehat{Y} \setminus \{0\}} \left| \mathbb{E}\left[e^{2\pi i \langle w, X \rangle} \mathbf{1}_{\{\|X\|^2 \ge K\}} \right] \right|^2. \tag{5.4}$$

The trivial character is omitted because R is centered. The task is to bound this whole sum, without multiplying a per-character bound by the enormous number N.

5.4 The smoothing estimates available before rejection

The reversed dual basis has Gram–Schmidt lengths reciprocal to the original ones, hence at least $1/B$. Summing its coefficients successively gives

$$\rho_c(\Lambda^*) \le \vartheta(c/B^2)^d, \qquad \vartheta(c) \le 1 + \sqrt{\pi/c}.$$

The first bound uses the fact that a shifted Gaussian sum is largest at an unshifted lattice. Poisson summation explains that fact: its expression for $\rho_c(L+z)$ has nonnegative Gaussian magnitudes multiplied by phases, so replacing every phase by one bounds it by $\rho_c(L)$.

Define $\eta_c = \vartheta(\pi^2/(cB^2))^d - 1$. For clarity, the Poisson identity used here is

$$\rho_c(L+z) = \frac{(\pi/c)^{d/2}}{\det L} \sum_{w \in L^*} e^{-\pi^2\|w\|^2/c} e^{2\pi i\langle w,z\rangle}.$$

Here $\det L$ is the volume of a basis parallelepiped; it equals N for a sublattice of $\mathbb{Z}^d$ of index N. The term $w = 0$ gives the central, translate-independent mass. The absolute sum of all other terms bounds its possible variation. This identity converts the geometric dual-mass estimate into the probability estimate that follows.

The same dual expansion yields

$$\frac{(\pi/c)^{d/2}}{N}(1-\eta_c) \le \rho_c(\Lambda + z) \le \frac{(\pi/c)^{d/2}}{N}(1+\eta_c) \tag{5.5}$$

for every real translate z. In particular $|p(y) - 1| \le \eta_\alpha$. When deriving this last bound, the integer dual vectors contribute phase one and must be canceled against the ambient normalization before taking absolute values. Counting them as error would weaken the bound.

An energy estimate is sharper than this pointwise bound. Put $A_0 = \pi^2/\alpha$. For a frequency coset $C = \mathbb{Z}^d - w$ and $0 < \epsilon < 2$, weighted Cauchy–Schwarz gives $\rho_{A_0}(C)^2 \le \rho_{\epsilon A_0}(C)\rho_{(2-\epsilon)A_0}(C)$. Apply it to each frequency coset, bound the first factor by an unshifted mass, and sum the second over the nontrivial characters. This gives

$$Q_0^2 \le \vartheta(\epsilon A_0)^d \left[\vartheta\left(\frac{(2-\epsilon)A_0}{B^2}\right)^d - 1\right]. \tag{5.6}$$

The certificate uses $\epsilon = 1/100$ here. The small tuning parameter ϵ in an inequality is independent of the width-design parameter $\varepsilon_{\mathrm{des}}$.

5.5 An exact integral for a discrete tail

Set $\beta = d/(2K)$ and $t = \alpha - \beta > 0$. The positive tilt t separates a Gaussian decay factor from an exact selector of integer norms. For $\operatorname{Re} z > 0$, define the twisted theta sum

$$\Theta(z,u) = \sum_{x \in \mathbb{Z}} e^{-zx^2 + 2\pi i u x}.$$

For each character w, its rejected-tail coefficient is exactly

$$\mathbb{E}\left[e^{2\pi i\langle w,X\rangle}\mathbf{1}_{\{\|X\|^2 \ge K\}}\right] \tag{5.7}$$

$$= \frac{e^{-tK}}{2\pi Z_\alpha^d} \int_{-\pi}^{\pi} \frac{\prod_{j=1}^d \Theta(\beta - i\theta, w_j) e^{-iK\theta}}{1 - e^{-t-i\theta}}\, d\theta. \tag{5.8}$$

To verify the selector, expand the denominator as $\sum_{j\ge 0} e^{-tj-ij\theta}$. A vector of squared norm m contributes a phase $e^{i(m-K-j)\theta}$, whose integral vanishes unless $j = m - K$. Such a nonnegative integer j exists exactly when $m \ge K$. The surviving weight is $e^{-tK}e^{-\beta m}e^{-t(m-K)} = e^{-\alpha m}$, as required. The positive β, t give absolute convergence, justifying the exchanges of integration and summation. This is an identity for the integer Gaussian, rather than a continuous radial approximation.

Minkowski's inequality allows us to bound the Euclidean norm of the vector of nontrivial coefficients by integrating its pointwise norm. We therefore need good aggregate estimates at the different angles θ.

5.6 Rational angles reveal the arithmetic cancellation

Write $\theta = \pi a/b + v$ with $\gcd(a, b) = 1$ and $b \ge 1$. The integer quadratic phase $\phi_{a,b}(x) = e^{i\pi a x^2/b}$ obeys $\phi_{a,b}(x + b) = (-1)^{ab}\phi_{a,b}(x)$. Let $\eta = 0$ when ab is even and $\eta = 1/2$ when it is odd. A finite Fourier expansion then has the form

$$\phi_{a,b}(x) = \sum_{\ell=0}^{b-1} c_\ell e^{2\pi i(\ell+\eta)x/b}, \qquad |c_\ell| = b^{-1/2}.$$

The half-frequency is necessary in the antiperiodic case. The uniform coefficient magnitude follows by autocorrelation: after removing the half-frequency, the correlation at shift h contains $\sum_{x=0}^{b-1} e^{2\pi i a h x/b}$, which vanishes unless b divides h. Fourier transforming this correlation gives $|c_\ell|^2 = 1/b$.

Put $h_v = \sqrt{\beta^2 + v^2}$ and $A_v = \pi^2\beta/(\beta^2 + v^2)$. The one-dimensional complex Poisson identity is

$$\Theta(z, u) = \sqrt{\frac{\pi}{z}} \sum_{k\in\mathbb{Z}} \exp\left(-\frac{\pi^2(k-u)^2}{z}\right), \qquad \operatorname{Re} z > 0, \quad u \in \mathbb{R}.$$

The square root is the branch positive on positive real z. For $z = \beta - iv$, its prefactor has modulus $\sqrt{\pi/h_v}$ and $\operatorname{Re}(1/z) = \beta/h_v^2$. The b shifted frequency sums from the finite expansion merge into $b^{-1}(\mathbb{Z} + \eta)$. Applying these facts in each coordinate gives

$$\left|\prod_{j=1}^{d} \Theta(\beta - i\theta, w_j)\right| \le \left(\frac{\pi}{bh_v}\right)^{d/2} \rho_{A_v}\Big(b^{-1}(\mathbb{Z}^d + \eta\mathbf{1}) - w\Big). \tag{5.9}$$

This converts oscillatory cancellation into a decaying Gaussian mass.

The decisive summation step uses the finite quotient. If $\gcd(b, N) = 1$, multiplication by b permutes $\widehat{Y}$. Consequently

$$\sum_{w\in\widehat{Y}} \rho_c\Big(b^{-1}(\mathbb{Z}^d + \eta\mathbf{1}) - w\Big) = \rho_{c/b^2}(\Lambda^* + \eta\mathbf{1}). \tag{5.10}$$

Indeed, multiplying every summation vector by b produces the sets $\mathbb{Z}^d + \eta\mathbf{1} - bw$. They form a disjoint decomposition of the translated dual lattice. The rescaling changes c to c/b^2. Thus all characters merge into one Gaussian sum, with no factor N. If coprimality fails, this equality acquires a kernel multiplicity and a smaller image lattice; dropping that correction is invalid.

Combining weighted Cauchy–Schwarz, this merger, the dual-mass estimate, and $Z_\alpha \ge \sqrt{\pi/\alpha}$ bounds the normalized Euclidean norm over all characters by

$$\left[\frac{\alpha}{bh_v}\vartheta\left(\frac{\epsilon A_v}{b^2}\right)\vartheta\left(\frac{(2-\epsilon)A_v}{b^2B^2}\right)\right]^{d/2}. \tag{5.11}$$

Omitting the trivial character can only decrease this norm.

5.7 Three angular regions, with a separate parity contribution

Set $Q_{\mathrm{arc}} = 512$ and $L = \pi/Q_{\mathrm{arc}}$. Every angle has a rational approximation with $1 \le b \le Q_{\mathrm{arc}}$ and $|v| \le \pi/(bQ_{\mathrm{arc}})$. All these denominators are coprime to $N = 12289^n$, because 12289 is prime and exceeds 512. Outside the regions within L of zero and the identified endpoints $\pm\pi$, one has $b \ge 2$. These remaining angles are called minor arcs.

For completeness, their continuous range reduces to an endpoint bound. Use $\epsilon = 1/2$, and define

$$r_{\min} = 2\beta, \quad r_{\max} = \sqrt{(Q_{\mathrm{arc}}\beta)^2 + (\pi/Q_{\mathrm{arc}})^2}, \quad u_0 = \sqrt{\frac{2}{\pi\beta}}, \quad v_0 = B\sqrt{\frac{2}{3\pi\beta}},$$
$$F(r) = \frac{\alpha}{r} + \alpha(u_0 + v_0) + \alpha u_0 v_0 r, \qquad C = \max\{F(r_{\min}), F(r_{\max})\}.$$

The approximation bounds imply $r = bh_v \in [r_{\min}, r_{\max}]$. The elementary theta estimate bounds the bracket in (5.11) by $F(r)$. Since $F''(r) = 2\alpha/r^3 > 0$, the maximum on this interval is attained at an endpoint. The contour denominator is at least $1 - e^{-t}$, so the entire minor-arc contribution satisfies

$$\overline{R}_{\mathrm{minor}} = \frac{e^{-tK} C^{d/2}}{1 - e^{-t}}.$$

No numerical sampling of the angular continuum enters this bound.

Near zero, the all-character estimate retains a large normalization term that the centered quantity R does not contain. Removing it gives

$$\sum_{w \ne 0} \rho_c(\mathbb{Z}^d - w) = \rho_c(\Lambda^*) - \rho_c(\mathbb{Z}^d) \le \vartheta(c/B^2)^d - 1.$$

The subtraction is the analytic counterpart of centering the density.

The other main region needs a different subtraction. At $\theta = \pi$, integer parity gives $(-1)^{\|x\|^2} = (-1)^{\sum_j x_j}$, the character shifted by $\mathbf{1}/2$. This contribution survives a sharp integer cutoff and would be missed by a continuous approximation. Since N is odd, $\mathbf{1}/2 \notin \Lambda^*$: otherwise the quotient dual would have an element of order two. Doubling injects $\Lambda^* + \mathbf{1}/2$ into $\Lambda^* \setminus \{0\}$, hence

$$\rho_c(\Lambda^* + \mathbf{1}/2) \le \rho_{c/4}(\Lambda^*) - 1 \le \vartheta(c/(4B^2))^d - 1.$$

Odd quotient order controls this parity term. Denominator coprimality controls the earlier merger; the two hypotheses play different roles.

To display the resulting integrals compactly, let

$$H_s(v;\epsilon) = \left(\frac{\alpha}{\sqrt{\beta^2 + v^2}}\right)^{d/2} \vartheta(\epsilon A_v)^{d/2} \sqrt{\vartheta\left(\frac{(2-\epsilon)A_v}{sB^2}\right)^d - 1}, \qquad s \in \{1, 4\}.$$

The two main contributions are bounded by

$$\overline{R}_0 = \frac{e^{-tK}}{\pi} \int_0^L \frac{H_1(v;\epsilon(v))}{\sqrt{(1-e^{-t})^2 + 4e^{-t}\sin^2(v/2)}}\, dv, \tag{5.12}$$

$$\overline{R}_\pi = \frac{e^{-tK}}{\pi(1 + e^{-t}\cos L)} \int_0^L H_4(v;\epsilon(v))\, dv. \tag{5.13}$$

The endpoint pieces at π and $-\pi$ together have length $2L$; both are included in the second expression. Each pointwise bound holds for every $0 < \epsilon < 2$, so its value may vary between integration subintervals. Minkowski's inequality now gives $R \le \overline{R}_0 + \overline{R}_\pi + \overline{R}_{\mathrm{minor}}$.

5.8 How a numerical enclosure completes the analytic proof

The remaining task is to evaluate explicit upper-bound expressions, not to sample keys or targets. Every real constant starts as an exact rational or decimal interval. Ball arithmetic encloses every operation by an interval, including rounding and input uncertainty.

Infinite theta sums have an explicit remainder. For every $x > 0$,

$$\vartheta(x) - 1 \le E(x) := 2\sum_{k=1}^{11} e^{-xk^2} + \frac{2e^{-144x}}{1 - e^{-25x}}.$$

For $k \ge 12$, consecutive summands have ratio at most e^{-25x}, which proves the geometric remainder. When $x < 1$, the identity $\vartheta(x) = \sqrt{\pi/x}\,\vartheta(\pi^2/x)$ moves the calculation to a rapidly convergent sum. Powers close to one are evaluated with $\log(1+u)$ and $e^u - 1$ routines that retain small corrections.

The main-arc integrals are enlarged to $[0, 512\beta]$. The certificate checks $L \le 512\beta < \pi$ and weakens the parity denominator accordingly. It divides $[0, \beta]$ and each of the nine dyadic blocks from $[\beta, 2\beta]$ to $[256\beta, 512\beta]$ into 512 pieces. There are 5120 upper rectangles per arc. On a piece $[\ell, h]$, the prefactor involving $\sqrt{\beta^2 + v^2}$ is largest at ℓ, the theta factors are largest at h, and the zero-arc denominator is smallest at ℓ. Multiplying these separate factor bounds and the piece width gives a proved upper rectangle. Monotonicity of their product is unnecessary. The choice is $\epsilon = 1/100$ in $[0, \beta]$ and $\epsilon = 1/2$ afterwards.

For orientation, the following entries are upward-rounded base-two logarithms of the certified upper-bound expressions.

Quantity	$n = 512$	$n = 1024$
Zero-arc bound	-68.25478	-84.41017
Parity-arc bound	-45.60045	-60.41745
Minor-arc bound	-222.19277	-470.14170
Unconditioned image norm bound $\overline{Q}_0$	-40.76750	-41.76000

The parity contribution dominates the tail estimate. The resulting conservative contracts are $R < 2^{-45.59}$ and $R < 2^{-60.41}$. A lower endpoint of an interval enclosing an upper-bound expression is not a lower bound on the actual R for an unknown key.

The certificate also checks $t > 0$, $\eta_\alpha < 1$, $D_0 < 1$, and $C < 1$ for the minor-arc envelope; D_0 is the uniform rejection bound derived next. These checks validate the positive denominators and decay factors used in the proof.

This computation uses 256-bit ball arithmetic, exact input intervals, explicit series remainders, and proved rectangle bounds. Its software trust includes the arithmetic implementation. Separate finite floating-point tests check phases and normalizations on small quotients; they are diagnostics, rather than proofs of these high-dimensional inequalities.

5.9 Turn tail energy into the two promised probability bounds

A uniform rejection bound follows from exponential Markov inequality and (5.5):

$$\delta_y \le D_0 := e^{-tK} \left(\frac{\alpha}{\beta}\right)^{d/2} \frac{1 + \eta_\beta}{1 - \eta_\alpha}. \tag{5.14}$$

Thus $a_y \geq 1 - D_0$ for every target, and $\delta \leq D_0$. The certified logarithms of this rejection bound are at most -14.31496 and -28.62992, respectively. These bounds are much larger than the acceptance variances because rejection and variation of rejection are different quantities.

Using $\delta_y = r(y)/p(y)$ gives $\delta_y - \delta = (r(y) - \delta p(y))/p(y)$. Since $p(y) \geq 1 - \eta_\alpha$, the triangle inequality yields

$$\mathrm{Var}_U(a_y) \leq \frac{(R + D_0 Q_0)^2}{(1 - \eta_\alpha)^2}. \tag{5.15}$$

The variance is no larger than the mean squared distance from any constant, so using the ambient mean δ is legitimate even though δ need not equal $\mathbb{E}_U \delta_y$.

Finally (5.3) implies

$$\frac{d\nu}{dU} - 1 = \frac{(p-1) - (r-\delta)}{1-\delta}, \qquad \chi^2(\nu \| U) \leq \frac{(Q_0 + R)^2}{(1 - D_0)^2}.$$

The certified variance expressions have logarithms below -91.19687 and -120.83204; the accepted-image expressions have logarithms below -81.43535 and -83.51999. These imply the simpler powers in (5.2). The accepted image is limited mainly by the original image error Q_0 at these bounds, so its two exponents need not improve at the same rate as the tail-variation exponents.

5.10 Individual coefficient caps preserve the stated bounds

The optional cap is a further predicate, so it needs its own estimate. Fix a coordinate set $I \subseteq \{1, \ldots, d\}$ with $|I| = m$. Completing the square in a single coordinate and using the translated coset bounds gives, for real s,

$$\mathbb{E}[e^{sX_j} \mid y] \leq \frac{1 + \eta_\alpha}{1 - \eta_\alpha} e^{\sigma^2 s^2 / 2}.$$

For a threshold $L_0 > 0$, optimizing the exponential tail bound and summing over the m coordinates in I gives

$$C_{\mathrm{cap}} := 2m \frac{1 + \eta_\alpha}{1 - \eta_\alpha} e^{-L_0^2/(2\sigma^2)} \;\geq\; \Pr[\exists j \in I : \; |X_j| \geq L_0 \mid y].$$

For $m = n$ and $L_0 = 2048$, these bounds are below $2^{-100.14568}$ and $2^{-95.70363}$. If b_y is the probability of passing the norm test but failing the cap, then $0 \leq b_y \leq C_{\mathrm{cap}}$ and the new acceptance probability is $a'_y = a_y - b_y$. The square root of its variance changes by at most $C_{\mathrm{cap}}/2$: for any variable in $[0, C_{\mathrm{cap}}]$, its variance is no greater than its mean squared distance from $C_{\mathrm{cap}}/2$, which is at most $C_{\mathrm{cap}}^2/4$. Thus, if V denotes the right-hand bound in (5.15),

$$\mathrm{Var}_U(a'_y) \leq (\sqrt{V} + C_{\mathrm{cap}}/2)^2.$$

Let ν' be the ambient syndrome law conditioned on passing both the norm test and the cap. Its accepted-image bound becomes

$$\chi^2(\nu' \| U) \leq \frac{\left(Q_0 + R + (1 + \eta_\alpha) C_{\mathrm{cap}}/2\right)^2}{(1 - D_0 - C_{\mathrm{cap}})^2}.$$

The additions fit within the stated powers of two. A cap on individual coordinates does not bound the sum of their encoded lengths; a retry triggered by a total padded byte budget remains a different problem.

The analytic conclusion is now in the form needed for security: uniform acceptance is bounded away from zero, the accepted ambient syndrome image has a tiny squared-density error, and these estimates hold for every lattice in the specified geometric class. Turning them into a signing-oracle comparison still requires the sampler profile, the direction of the distribution comparison, and the oracle reduction.

6 From a sampled vector to signature bytes

The distributional theorem studies a full pair (u, v). A public signature transmits the salt and v, from which the verifier reconstructs the centered component $\widehat{u}$. This deterministic map preserves validity: the syndrome is unchanged and $\|(\widehat{u}, v)\| \le \|(u, v)\|$. The distinction between the sampled pair and its representation also matters for the strongest forgery definition.

6.1 Coefficient limits and buffer capacity

In the canonical compressed format, each coefficient must satisfy $|v_i| \le 2047$. The optional coordinate-cap theorem gives a distributional estimate when that restriction is included in acceptance. The norm-only signer can instead sample as before and return a public error if the accepted vector cannot be encoded. These are two explicitly different ways to specify a signer.

An encodable coefficient uses a sign bit, seven low bits, and a unary high part. Its length is

$$9 + \left\lfloor \frac{|v_i|}{128} \right\rfloor \quad \text{bits.}$$

For a norm-passing vector, Cauchy–Schwarz gives

$$\sum_{i=1}^{n} |v_i| \le \sqrt{n \sum_{i=1}^{n} v_i^2} \le \sqrt{nT}.$$

Since a sum of floors is no greater than the floor of the sum, a header of one byte and the 40-byte salt give the total bound

$$41 + \left\lceil \frac{9n + \left\lfloor \sqrt{nT}/128 \right\rfloor}{8} \right\rceil \quad \text{bytes.} \tag{6.1}$$

The ceiling converts bits to whole bytes. The resulting values are 746 and 1455 bytes for degrees 512 and 1024. They fit in the specified maximum compressed buffers of 752 and 1462 bytes. This argument assumes coefficient encodability; the norm bound by itself does not guarantee that every coefficient satisfies the codec cap.

A smaller fixed padded byte budget imposes another restriction on the sum of encoded lengths. Repeating the sampling loop until that budget is met changes the acceptance event. The individual coefficient-cap estimate does not supply the corresponding distributional theorem.

6.2 Why a unique representation matters for forgery

Under existential unforgeability with chosen-message access (EUF), a forgery must concern a message never requested, including requests that returned a public error. Strong unforgeability (SUF) requires a valid message/signature pair never successfully returned, even for a requested message. It can involve an earlier salt and hash input. For either notion F, write $\mathrm{Adv}^{\mathsf{F}}(\mathcal{A})$ for adversary $\mathcal{A}$'s probability of producing a successful forgery.

The compressed wrapper accepts one canonical format and has enough space for every norm-passing, coefficient-encodable vector. A new canonical signature at the same input must contain a different v. If the earlier request returned an encoding error, its hidden v has no admitted encoding, so any valid forgery again differs. The reconstructed pair is a different preimage regardless of centering.

Sufficient space, a unique format, and the stated error semantics enter the theorem. Multiple byte strings for one v, or errors caused by undersized buffers, require a separate extraction argument.

7 Turning the distribution estimate into security

The analytic result concerns Gaussian vectors and their syndromes. A signature theorem concerns an adversary choosing messages, seeing signatures, and trying to produce a new valid signature. The bridge between them is a public simulator: it generates short vectors without the secret key, then arranges that the hash values agree with their syndromes. We must quantify both the change of distribution and the cost of arranging those hash values.

7.1 A moment measures the cost of changing an experiment

For probability laws P, Q on a countable set and a real order $\rho > 1$, define the power moment

$$M_\rho(P\|Q) = \sum_x P(x)^\rho Q(x)^{1-\rho}. \tag{7.1}$$

It is finite only when Q gives positive probability wherever P does. It equals one when the laws agree, and is always at least one. The corresponding multiplicative divergence is $M_\rho^{1/(\rho-1)}$; its natural logarithm is the Rényi divergence. We use the unrooted moment throughout, so that products of independent experiments multiply their moments. This order ρ is unrelated to an exponent used in the norm-tail contour.

For any event E, Hölder's inequality gives

$$P(E) \le M_\rho(P\|Q)^{1/\rho} Q(E)^{(\rho-1)/\rho}. \tag{7.2}$$

Indeed, write $P(E) = \sum_x Q(x)(P(x)/Q(x))\mathbf{1}_E(x)$ and use conjugate exponents ρ and $\rho/(\rho-1)$. The same argument applies to any success function taking values in $[0,1]$. This is how a bound for one simulation becomes a bound for the real adversary's success.

7.2 Rejection should preserve a small error, not create one

Recall that P_y is the declared exact-arithmetic tree-sampler law, while G_y is the exact coset Gaussian. Suppose a proven sampler profile gives

$$\sup_y \{M_\rho(P_y\|G_y) - 1\} \le e_p(\rho), \qquad P_y(z)/G_y(z) \le b. \tag{7.3}$$

The norm-retry loop conditions each law on the acceptance set A. A crude division by the acceptance probability would charge a loss even if the sampler were already exact. The correct quantity to propagate is the *excess above one*.

Let $s = P(E) > 0$ and $t = Q(E) > 0$. Splitting the moment between E and its complement gives

$$\begin{aligned} M_\rho(P\|Q) = {} & s^\rho t^{1-\rho} M_\rho(P \mid E \| Q \mid E) \\ & + (1-s)^\rho (1-t)^{1-\rho} M_\rho(P \mid E^c \| Q \mid E^c). \end{aligned} \tag{7.4}$$

Each conditional moment is at least one. The sum obtained by replacing both conditional moments by one is itself a probability moment, for the two Bernoulli laws with parameters s, t, and is at least one. Consequently

$$M_\rho(P \mid E \| Q \mid E) - 1 \le \frac{t^{\rho-1}}{s^\rho}\{M_\rho(P\|Q) - 1\}. \tag{7.5}$$

Zero-probability complementary events are omitted or handled by continuity. When $P = Q$, the right-hand side is zero, as it should be.

The uniform rejection estimate from the analytic argument is $d_y = G_y(A^c) \le D_0$. The likelihood-ratio upper bound gives $P_y(A) \ge 1 - bd_y$. Assuming $bD_0 < 1$, the increasing function $(1 - x)^{\rho-1}/(1 - bx)^\rho$ therefore yields

$$M_\rho(P_y \mid A \| G_y \mid A) - 1 \le C_\rho e_p(\rho), \qquad C_\rho = \frac{(1 - D_0)^{\rho-1}}{(1 - bD_0)^\rho}. \tag{7.6}$$

For b close to one, C_ρ is close to $1/(1 - D_0)$. The norm test thus amplifies an existing small sampler discrepancy; it does not create a standalone discrepancy of the size of the rejection probability.

For the specified exact-arithmetic sampler, the width-design parameter ε_{des} is $2^{-35.5}$ or 2^{-36} at the two degrees, and $d = 2n$. Its probability-ratio contract is

$$a \le P_y(z)/G_y(z) \le b, \qquad b = \left(\frac{1 + \varepsilon_{\text{des}}/d}{1 - \varepsilon_{\text{des}}/d}\right)^d, \qquad a = b^{-1}. \tag{7.7}$$

This imports the tree sampler's actual normalization-product guarantee; it is not inferred by identifying it with an arbitrarily ordered lattice sampler. Under G_y, the ratio $W = P_y/G_y$ has mean one. Since $(W - a)(b - W) \ge 0$, its variance is at most $(b - 1)(1 - a)$. A second-order Taylor bound for x^ρ at one gives the nonasymptotic profile

$$e_p(\rho) = \frac{\rho(\rho - 1)}{2} \max\{a^{\rho-2}, b^{\rho-2}\}(b - 1)(1 - a). \tag{7.8}$$

For $\rho \ge 2$, the maximum is $b^{\rho-2}$. A different implementation could enter the abstract proof through its own proven profile, including its finite-precision effects.

7.3 Why the direction of the distribution comparison matters

Let ν be the syndrome law of the accepted ambient Gaussian, and put $w(y) = |Y|\nu(y)$. Recall the densities p and r for the whole Gaussian image and its rejected part, and the ambient rejection probability δ. Then

$$w(y) = \frac{p(y) - r(y)}{1 - \delta}, \qquad w(y) - 1 = \frac{(p(y) - 1) - (r(y) - \delta)}{1 - \delta}. \tag{7.9}$$

The certified bounds $\overline{Q}_0, \overline{R}$ imply

$$\mathbb{E}_U(w - 1)^2 \le \frac{(\overline{Q}_0 + \overline{R})^2}{(1 - D_0)^2} =: X_0. \tag{7.10}$$

The same proof supplies $p(y) \ge 1 - \eta > 0$, where $\eta = \eta_\alpha$, and $G_y(A) \ge 1 - D_0$, hence

$$w(y) = \frac{p(y)G_y(A)}{1 - \delta} \ge (1 - \eta)(1 - D_0) =: m_0 > 0. \tag{7.11}$$

The desired transfer is from a uniform real target to the simulator's target, so it needs $M_\rho(U\|\nu)$, the reverse direction from the forward chi-square estimate $\mathbb{E}_U(w - 1)^2$. A tiny set on which w nearly vanishes could make a reverse higher-order moment large. The density floor rules out that possibility.

In fact, for $f(x) = x^{1-\rho}$, Taylor's theorem gives $f''(x) = \rho(\rho - 1)x^{-\rho-1}$. Between 1 and $w(y)$ this is at most $\rho(\rho - 1)m_0^{-\rho-1}$. The linear term averages to zero because $\mathbb{E}_U w = 1$, so

$$M_\rho(U\|\nu) - 1 = \mathbb{E}_U w^{1-\rho} - 1 \le \frac{\rho(\rho - 1)}{2} m_0^{-\rho-1} X_0 =: e_u(\rho). \tag{7.12}$$

Both the untruncated image error and the norm-tail error enter X_0. The variance of the acceptance probabilities alone would describe a different target experiment.

7.4 One joint change handles adaptive messages

After isolating each fresh signing input in the random oracle, the real accepted pair and the public pair have laws

$$L(y,z) = U(y)P_y(z \mid A), \qquad J(y,z) = \nu(y)G_y(z \mid A). \tag{7.13}$$

The public law can be sampled without a trapdoor: draw z from $G \mid A$ and compute $y = F_h(z)$. Combining the target and conditional-preimage moments in one sum gives

$$M_\rho(L\|J) \le (1 + C_\rho e_p(\rho))(1 + e_u(\rho)) =: M_*. \tag{7.14}$$

Keeping the pair together matters: the final event bound pays one exponent $(\rho-1)/\rho$ for this whole distribution change.

Fix a key. In the resampled signing game, each pair law depends on that key but not on the chosen message. Let q_S bound the number of signing requests. We may therefore prepare q_S independent pairs before the adversary starts. A common interaction procedure consumes one pair per signing request, draws the fresh salt, associates its target with the currently chosen message, programs the oracle, and returns the prescribed signature representation. All message adaptivity is inside that same procedure. The input lists have moment at most $M_*^{q_S}$.

For each fixed list, even a quantum interaction has a final success probability in $[0,1]$. Applying (7.2) to that success function proves

$$\Pr_L[\text{forge}] \le M_*^{q_S/\rho} \Pr_J[\text{forge}]^{(\rho-1)/\rho}. \tag{7.15}$$

Uniformity of the analytic and sampler bounds permits averaging over key generation. Only the classical pair list changes; the common interaction procedure, including its oracle access, stays fixed. Signing requests remain classical; any initial quantum advice is independent of the oracle, and oracle-dependent preparation is charged to the adversary's query count.

7.5 The public simulator has a finite sampling budget

An unlimited retry loop is convenient for specifying a probability law, but a hardness reduction needs a resource bound. Draw $C \ge q_S$ independent vectors from the unconditioned ambient Gaussian, retain the first q_S norm-passing vectors, and abort if there are too few. With

$$\varepsilon_{\text{pool}} = \Pr[\text{Bin}(C, D_0) > C - q_S], \tag{7.16}$$

the abort probability is at most $\varepsilon_{\text{pool}}$. Here $\text{Bin}(C, D_0)$ counts failures in C independent trials with failure probability D_0; stochastic domination uses $\delta \le D_0$.

Conditional on any successful Boolean acceptance pattern, the retained values are independent with law $G \mid A$. Their joint law does not depend on which successful pattern occurred. If π_C is the actual pool-success probability, the finite simulator's forging probability is thus exactly π_C times the unlimited simulator's probability. We may upper-bound the latter by dividing the former by $1 - \varepsilon_{\text{pool}}$.

At $q_S = 2^{64}$, the certified choices are

$$C = 2^{64} + 2^{50} \quad (n = 512), \qquad C = 2^{64} + 2^{36} \quad (n = 1024), \tag{7.17}$$

and give $\varepsilon_{\text{pool}} < 2^{-1000}$. The binomial bound starts at the first failing integer count $C - q_S + 1$. The simulator uses at most Cd one-dimensional Gaussian draws and q_S syndrome computations. It makes only q_S oracle programs, since rejected pool vectors are never programmed. The sampling primitive retains its declared expected running time; a bound on its call count is not itself an implementation proof.

7.6 Which hardness assumptions receive a forgery?

Both lattice games use the same honestly generated public-key distribution as the signer, with the secret key discarded. They do not assume that the public ring element is uniform.

In the t-target *inhomogeneous short-preimage game*, labeled NTRU-ISIS here, the solver receives h and independent uniform syndromes $c_1, \ldots, c_t$. It must return an index j and (u, v) satisfying

$$u + hv = c_j \pmod q, \qquad \|(u, v)\|^2 \le T. \tag{7.18}$$

Denote its success probability by $\mathrm{Adv}^{t\text{-ISIS}}(B_{\mathrm{I}})$.

The additional strong-unforgeability game, labeled NTRU-SPISIS*, supplies C independent *unconditioned* vectors $z_i = (u_i, v_i) \leftarrow G$, together with their syndromes c_i. The solver sees all the triples and must return a short preimage of some c_j different from the supplied z_j. Write its advantage as $\mathrm{Adv}^{C\text{-SPISIS}^*}(B_{\mathrm{S}})$. The star records the unconditioned challenge distribution: a Gaussian pool conditioned in advance on passing its norms would be a different assumption.

A forgery at an unprogrammed hash input yields an ISIS solution. At a programmed input, a new signature yields a different preimage of one of the programmed pool syndromes. This second case is relevant for strong unforgeability, which forbids a new signature even on an already signed message. Ordinary unforgeability requires an entirely new message, so its forged input cannot have been programmed by a signing request.

The reconstruction and canonical-format conditions proved in Section 6 ensure that a new signature at a programmed input has a different transmitted v, including after an encoding error. It therefore yields a distinct reconstructed preimage, as the second game requires.

7.7 The classical random-oracle theorem

In the classical random-oracle model (ROM), let q_H bound hash queries and κ be the salt length in bits. On the jth signing request, at most $q_H + j - 1$ relevant inputs have already been entered into the oracle table. Independence of the fresh salt therefore bounds the total reprogramming discrepancy by

$$\varepsilon_{\mathrm{repr}}^{\mathrm{ROM}} = \frac{q_S q_H + \binom{q_S}{2}}{2^{\kappa}}. \tag{7.19}$$

Subsequent hash queries may depend on the revealed salt; the estimate only needs its freshness when the signing input is selected.

Under the uniform geometric and sampler hypotheses established above, fix $\rho > 1$ with $bD_0 < 1$, a pool size $C \ge q_S$, and $\varepsilon_{\mathrm{pool}} < 1$. For every classical adversary $\mathcal{A}$ within the query budgets, there exist corresponding hardness solvers B_{I} and, for SUF, B_{S}. Define their contributions by

$$H_{\mathrm{EUF}}^{\mathrm{ROM}} := \mathrm{Adv}^{(q_H+1)\text{-ISIS}}(B_{\mathrm{I}}), \tag{7.20}$$

$$H_{\mathrm{SUF}}^{\mathrm{ROM}} := H_{\mathrm{EUF}}^{\mathrm{ROM}} + \mathrm{Adv}^{C\text{-SPISIS}^*}(B_{\mathrm{S}}). \tag{7.21}$$

For either freshness definition, the resulting theorem is

$$\mathrm{Adv}^{\mathsf{F}}(\mathcal{A}) \le \varepsilon_{\mathrm{repr}}^{\mathrm{ROM}} + M_*^{q_S/\rho} \left(\frac{H_{\mathsf{F}}^{\mathrm{ROM}}}{1 - \varepsilon_{\mathrm{pool}}} \right)^{(\rho-1)/\rho}, \qquad \mathsf{F} \in \{\mathrm{EUF}, \mathrm{SUF}\}. \tag{7.22}$$

The ISIS solver assigns independent challenge targets to new unprogrammed hash inputs, reserving one additional target for the final forgery if necessary. The strong case uses the unconditioned challenge vectors as its public pool. These extractors run the adversary once; splitting the two disjoint forgery cases gives the sum above.

Each reduction pays for the Gaussian pool, syndrome arithmetic and a classical table of at most $q_H + q_S + 1$ entries, together with challenge reading and ordinary bookkeeping. Its work is not simply the adversary's work. A numerical lattice-hardness estimate must be evaluated at those actual resources. Every probability bound can of course be capped at one.

7.8 Quantum hash queries change the oracle costs

In the quantum random-oracle model (QROM), signing is still classical, but hash queries may be superpositions of inputs. The retained adaptive programming bound is

$$\varepsilon_{\mathrm{repr}}^{\mathrm{QROM}} \le \frac{3q_S}{2}\sqrt{\frac{q_H + q_S + 1}{2^\kappa}}. \tag{7.23}$$

Private norm retries add no programs or hash queries: the target was obtained once at the start of the signing request.

For extraction, implement the simulated oracle as a static background function H_0 and a mutable classical override table. On first programming an input, query and store its original H_0 value and its desired current value. A quantum query applies H_0, then reversibly XORs the bitwise difference of the original and current encodings at matching table entries. Here XOR means addition of corresponding bits modulo two. Repeated programs update the desired value while preserving the original. Including signing programs and final verification, the number of background queries is at most

$$Q = q_H + q_S + 1. \tag{7.24}$$

The table can be compiled into an ordinary reversible circuit. A direct implementation scans up to q_S entries per query; it requires no quantum random-access-memory instruction, but its gate overhead is not constant.

Quantum extraction measures a candidate oracle input and reprograms it to a supplied uniform challenge, with a loss bounded by $(2Q+1)^2$. The success predicate includes absence of that input from the eventual override domain. A point later programmed by signing therefore fails the predicate, rather than being silently treated as untouched. Retain also the conservative target-value collision allowance

$$\varepsilon_{\mathrm{hit}} = \frac{10Q^2 q_S}{|Y|}. \tag{7.25}$$

The programmed target list is independent of the background oracle, as required by that search allowance. Put

$$H_{\mathrm{EUF}}^{\mathrm{QROM}} := (2Q+1)^2\,\mathrm{Adv}^{\text{1-ISIS}}(B_{\mathrm{I}}) + \varepsilon_{\mathrm{hit}}, \tag{7.26}$$

$$H_{\mathrm{SUF}}^{\mathrm{QROM}} := H_{\mathrm{EUF}}^{\mathrm{QROM}} + \mathrm{Adv}^{C\text{-SPISIS}^*}(B_{\mathrm{S}}). \tag{7.27}$$

For every quantum adversary within the stated budgets, there are corresponding quantum hardness solvers for which (7.22) holds with ROM replaced by QROM in its programming and hardness terms. The other geometric, sampler, and pool hypotheses remain in force. The extraction solver has polynomial local simulation overhead and the finite Gaussian-pool cost. Eliminating the ideal background oracle uses a sufficiently independent family evaluated reversibly; this does not establish near-linear runtime in Q. The stated corollary uses single-target extraction and no quantum random-access-memory assumption.

7.9 Interpreting the concrete losses

Suppose the unlimited public simulator's forging probability is at most $2^{-\lambda_{\mathrm{sim}}}$. Equation (7.15) then gives a real-experiment exponent at least $\lambda_{\mathrm{sim}} - \mathcal{L}(\rho)$, where the exact transfer loss is

$$\mathcal{L}(\rho) = \frac{\lambda_{\mathrm{sim}}}{\rho} + \frac{q_S}{\rho} \log_2 M_*. \tag{7.28}$$

Rigorous interval evaluation at $q_S = 2^{64}$ gives the following outward rounded bounds. The logarithm is base two, so the unit of loss is bits.

n	λ_{sim}	ρ	$\mathcal{L}(\rho)$
512	120	73	< 3.268
512	128	75	< 3.376
1024	256	151	< 3.387

In these rows, the accepted-image contribution is respectively below 0.000398, 0.000409 and 0.000207 bits. The additional sampler cost caused by conditioning is below 0.000080, 0.000082 and $5 \cdot 10^{-9}$ bits. These figures isolate a distribution transfer; they are not measured or proven lattice work factors.

The other costs remain visible. At $q_S = 2^{64}$ and $\kappa = 320$, the classical programming expression is approximately $2^{-191.415}$ for $q_H = 2^{64}$ and 2^{-128} for $q_H = 2^{128}$. The conservative quantum expression is instead approximately $2^{-62.915}$ and $2^{-31.415}$. These are separate additive probability terms, before accounting for the quantum extraction loss and simulation work. A roughly 3.3-bit moment loss cannot remove them. This is why the classical theorem is the principal concrete-security application.

Finally, an optional real signer capped at ℓ norm attempts per query adds at most $q_S(bD_0)^\ell$ to either theorem. At $q_S \le 2^{64}$, $\ell = 32$ for degree 512 and $\ell = 16$ for degree 1024 give a bound below 2^{-390}. This is a different failure event from exhaustion of the simulator's public Gaussian pool. Neither estimate covers a revealed retry count or timing transcript: the modeled signing interface exposes only the final signature or its specified public error.

8 What the complete argument tells us

The proof connects two experiments on the same public equation. The signer chooses a target and samples an accepted preimage; the public simulator chooses an accepted vector and computes its target. The norm cutoff introduces target-dependent acceptance weights, so ordinary Gaussian smoothing must be supplemented by control of the rejected image.

The key-quality bound controls dual-lattice Gaussian masses. The exact integer-tail contour separates zero, parity, and remaining rational-angle contributions. Merging the character classes avoids a factor equal to the number of syndromes. Certified evaluation then bounds the acceptance variance and accepted-image chi-square distance for every permitted key.

The security argument preserves these small errors through conditioning and interaction. The conditional-excess lemma handles the sampler; the density floor supplies the reverse target moment. Comparing target and preimage jointly gives one event-probability exponent, and a common procedure consumes independent pairs even when messages are chosen adaptively.

Fresh salts and a finite public Gaussian pool complete the simulation, with an explicit programming error, sampling-call budget, and pool-failure bound. Unprogrammed-input forgeries yield

the uniform-target problem; strong forgeries additionally use the distinct-preimage game and canonical representation. The quantum-oracle corollary retains its programming, extraction, and simulation costs.

8.1 How to interpret the quantitative result

There are several different quantities in this chain, and their units matter. The variance is a squared-average variation in acceptance probability over uniform targets. The chi-square distance is a squared density discrepancy of the accepted public image. Both are dimensionless. The transfer loss is a base-two logarithmic loss in a conditional forgery-probability bound, measured in bits. Signing and hash queries are counts of oracle interactions. The Gaussian-pool budget counts sampler calls, while the reduction's local operations contribute actual work. None of these quantities can replace the others in the final theorem.

The rejection bound and the assumptions behind the bit-loss calculation must therefore remain attached to those results when interpreting them.

The theorem applies to every key meeting its geometric hypotheses. The cryptographic assumptions, however, are evaluated on the actual specified trapdoor-generated key distribution. Uniform analytic control does not prove hardness for that distribution, and the numerical certificate does not estimate an attacker's work factor.

8.2 The model boundary is part of the result

The covered signer draws one independent salt after each chosen message, keeps its target fixed during private norm retries, uses a sampler with the declared distributional profile, and returns only the final specified output. The concrete sampler application is exact arithmetic. The proof covers the mathematical signature and the stated canonical compressed wrapper with adequate buffers and public encoding errors.

A finite-precision signer requires a profile for its actual sampling errors. Rejection on a total padded encoding budget introduces another conditioning event. Caller-selected or reused salts change the fresh-input analysis. Exposed retry counts or timing provide a different observation to the adversary. Several keys or an aggregate interface require corresponding key and oracle models. Each change identifies a definite additional proof obligation.

A Notation at a glance

The table collects the recurring objects. Gaussian exponents and probability-comparison orders use different symbols because they perform different jobs.

Symbol	Meaning
n, $d = 2n$	Polynomial degree and dimension of the integer coefficient vector.
$q = 12289$	Modulus in the public syndrome equation.
F_h, Λ_h	Public map $(u, v) \mapsto u + hv \bmod q$, and its kernel lattice.
Y, $N = \lvert Y \rvert$	Syndrome quotient and its cardinality q^n.
U	Uniform probability law on the finite quotient.
σ, α	Gaussian width and exponent parameter $\alpha = 1/(2\sigma^2)$.
$\rho_c(S)$	Unnormalized Gaussian mass $\sum_{x \in S} e^{-c\lVert x \rVert^2}$, for $c > 0$.
G, G_y	Ambient integer Gaussian and that Gaussian conditioned on syndrome y.
P_y	Actual exact-arithmetic preimage-sampler law under the declared contract.
T, $K = T + 1$	Largest accepted squared norm and first rejected integer squared norm.
A	Event $\lVert z \rVert^2 \le T$.
a_y	Exact coset-Gaussian acceptance probability $G_y(A)$.
$p(y)$, $r(y)$	Unconditioned and rejected ambient-image densities relative to U.
δ	Ambient rejection probability $\mathbb{E}_U r$; not generally the uniform-coset mean rejection probability.
Q_0, R	Centered image and tail density norms $\lVert p - 1 \rVert_{L^2(U)}$ and $\lVert r - \delta \rVert_{L^2(U)}$.
D_0	Uniform upper bound on rejection probability in any coset.
ν, $w(y)$	Accepted ambient syndrome law and its density $w = \mathrm{d}\nu/\mathrm{d}U$.
$\rho > 1$, M_ρ	Order and unrooted likelihood-ratio moment used in the security comparison.
L, J	Joint target/preimage laws for fixed uniform-target signing and public accepted-Gaussian simulation.
q_S, q_H, κ	Signing queries, hash queries, and salt length in bits.
C, $\varepsilon_{\mathrm{pool}}$	Finite Gaussian-pool size and upper bound on its failure probability.
$Q = q_H + q_S + 1$	Background-oracle query budget in the stated quantum reduction.